

15. (b) Once accepted, zero became a foundation for higher mathematics and modern technology.
16. (d) SQPR
S: of learning throughout the ages,
Q: not indeed, in the sense that education has been universal
P: but in the sense that learning has always been very highly valued
R: and the learned man has been held in higher esteem than the warrior or the administrator.
Sentence: *India has been a land of learning throughout the ages, not indeed in the sense that education has been universal, but in the sense that learning has always been very highly valued, and the learned man has been held in higher esteem than the warrior or the administrator.*
17. (a) PSRQ
P: century show that the June rainfall is
S: no predictor of the monsoon's outcome
R: rain may have soothed concerns
Q: of a drought, India's rainfall data for over a
Sentence: *While the recent rain may have soothed concerns of a drought, India's rainfall data for over a century show that the June rainfall is no predictor of the monsoon's outcome.*
18. (a) SRQP
S: means hatred for other countries, but this
R: type of patriotism is an evil and it
Q: should be condemned because
P: today man belongs to the whole world as much as to any one country.
Sentence: *For some people patriotism means hatred for other countries, but this type of patriotism is an evil and it should be condemned because today man belongs to the whole world as much as to any one country.*
19. (b) Bee produces honey; Cow produces milk.
20. (a) Oven is used to bake; Knife is used to slice.
21. (c) Finger is a part of a hand; Leaf is a part of a tree.
22. (b) Rule: When we refer to a distance, period of time, or amount, it is considered singular.
23. (b) Rule: *Neither / Either / Each* are **singular**.
Incorrect: *Neither of the students have completed...*
Correct: *Neither of the students has completed...*
24. (b) Rule: In indirect sentences after "know," the structure should be **to whom she should** (not "to whom should she").
25. (a) Impartial → fair, unbiased → correct word for a judge.
Biased, emotional, hesitant → inappropriate.
26. (c) Rule: *Hardly / Scarcely* ... when → always takes **past perfect tense**.
27. (a) *Came across* → to find by chance (correct usage).
Came by → to obtain something (less suitable here).
Came out → to be published / revealed.
Came into → to inherit.
28. (d) Present tense is required here because it's a general fact.
29. (b) The word "tolerance" fits well with "adverse conditions."
Other options don't suit the meaning.
30. (b) If a plant lacks drought resistance, it is artificially **irrigated** (given water).
Other options (heated, showered, deserted) do not fit the context.
31. (b) The language used in the most inscriptions of Ashoka is Prakrit.
Some inscriptions in the northwestern regions were also in **Greek** and **Aramaic**.
Prakrit was usually in the **Brahmi script** in most of India and the **Kharosthi script** in the northwest.
Ashoka wanted his messages of **dhamma** to reach the **common people**, so he used **Prakrit**, not Sanskrit or Pali.
32. (c) The 'NIDHI' scheme was launched by department of Science and Technology in 2016.
NIDHI is designed to nurture technology-driven ideas into startups and encourages incubation, and supports academic and R&D institutions in fostering innovation through incubation infrastructure.
The acronym NIDHI stands for **National Initiative for Developing and Harnessing Innovations**.
NIDHI is **implemented by the Department of Science & Technology (DST)** via its Technology Translation and Innovation Division and the National Science & Technology Entrepreneurship Development Board (NSTEDB). So, statements 2, 3, 4 are correct.
33. (a) The RailOne App, launched by Indian Railways in 2025, primarily aims to provide passengers with a single-window platform for integrated services.
It brings together age-old features like reserved and unreserved ticket booking, PNR status, live train tracking, food ordering, grievance redressal, and more—all under one roof.
It also allows single-sign-on login using mPIN or biometric methods.
34. (b) General Bipin Rawat, the first Chief of Defence Staff (CDS) of India, was commissioned into Gorkha Rifles.
He Held **counter-insurgency** commands in Jammu & Kashmir and the Northeast.
He is known for modernization efforts, jointness among services, and strong stance on national security.
35. (c) The primary cause of acid rain formation is sulfur dioxide and nitrogen oxides emissions.
The gases are released into the atmosphere mainly from burning coal, oil, vehicle exhausts, and industrial processes.
These gases react with **water vapour, oxygen, and other chemicals** in the atmosphere to form **sulfuric acid** and **nitric acid**.
When this acidic mixture falls as rain, snow, or fog, it is called **acid rain**.
36. (c) Banu Mushtaq was awarded the International Booker Prize for the year 2025.
She is a distinguished Kannada-language writer, who was awarded the prize for her short story collection "Heart Lamp".
'Heart Lamp' made history as both the **first short story collection** and the **first work translated from Kannada** to win the International Booker Prize.
37. (c) Exercise PRALAY was carried out by Indian Air Force.
It was a **command-level exercise** aimed at testing combat readiness and jointness with the Indian Army.
The drill involved **fighter aircraft, transport aircraft, helicopters, and missile systems** operating from forward bases near the Line of Actual Control (LAC).
Its focus was on **high-intensity combat operations, rapid deployment, and operational preparedness** in sensitive border areas.
38. (b) Project 75, under which India is constructing Scorpene-class submarines, is being executed by Mazagon Dock Shipbuilders Limited.
Project 75 is a program to build **six Scorpene-class diesel-electric submarines** in India.
These submarines are being constructed in collaboration with **Naval Group of France**.
So far, submarines like **INS Kalvari, INS Khanderi, INS Karanj, INS Vela, and INS Vagir** have already been commissioned into the Indian Navy.

39. (d) Falcon 9 Block 5 is used to launch the Axiom Mission 4 (Ax-4) to the International Space Station.

Axiom Mission 4 (Ax-4) was launched on June 25, 2025 from Launch Complex 39A at NASA's Kennedy Space Center.

The mission carried a **SpaceX Crew Dragon capsule named "Grace"** and transported four astronauts to the International Space Station.

40. (d) The purpose of 'Operation Sindhu' was to safely evacuate Indians stranded in Iran.

Operation Sindhu was launched on **18 June 2025** by the Government of as a **humanitarian evacuation mission**.

Its primary purpose was to **evacuate Indian nationals—especially students—stranded in Iran amidst the escalating Iran–Israel conflict**.

The operation has successfully evacuated thousands through multiple phases, including routes via Armenia.

41. (b) Sir Alexander Cunningham established the Archaeological Survey of India (ASI).

Cunningham was appointed as the first **Director-General of ASI**.

His work included surveying ancient monuments, excavating historical sites, and preserving India's cultural heritage.

42. (d) The **Reserve Bank of India (RBI)** launched the **DAKSH portal** in **November 2022**.

The RBI supervises thousands of **banks, NBFCs, payment system operators, etc.**, and these entities must regularly **report compliance, respond to inspections, and submit regulatory data**.

DAKSH streamlines this by acting as a **single, online workflow platform**.

Thus, The correct statements are 1, 2, and 3.

43. (b) The primary aim of the "100 Million for 100 Million" campaign launched in 2016 is to mobilize 100 million youth to support 100 million children denied basic rights and freedom.

The focus was on **ending child labour, child slavery, violence, and exploitation**, while ensuring access to education and dignity.

This was a **global youth-led movement** under the **Kailash Satyarthi Children's Foundation**.

44. (a) Ganesh Vasudev Mavalankar is known as the "Father of the Lok Sabha".

He was the **first Speaker of the Lok Sabha**, serving from **1952 to 1956**.

Earlier, he was also the **Speaker of the Central Legislative Assembly** and the **Constituent**

Assembly (Legislative) before becoming Speaker of the independent India's **first Lok Sabha**.

45. (a) The working principle of washing machine is centrifugation.

When the drum rotates at high speed, **centrifugal force** pushes water and dirt outward through the holes of the drum, while clothes remain inside.

This process helps in **separating water from wet clothes**.

46. (b) Cumulonimbus clouds are typically associated with Thunderstorms.

They form due to strong **upward convection currents** in the atmosphere.

Weather associated with cumulonimbus clouds includes **Thunderstorms, heavy rain, and lightning**.

47. (d) The position of the Indian football team in the 1956 Melbourne Olympics was 4th.

India reached the **semi-finals** after defeating **Australia 4–2**.

In the **semi-final**, India lost to **Yugoslavia (1–4)**.

In the **bronze medal match**, India lost to **Bulgaria (0–3)**.

48. (b) Statements 1 and 3 are correct.

Isotopes have the same **proton number**, but different **neutron numbers**.

In **isotones**, **number of protons** is different, while **neutrons** are same.

Unlike isotopes, isotones belong to **different elements**.

49. (c) The Pechora surface-to-air missile system currently in service with the Indian Air Force, was originally developed by Russia (former Soviet Union) in the 1960s.

The **Pechora system** is a **medium-range surface-to-air missile (SAM)**.

India inducted it in the **1970s**, and it still serves in the **Indian Air Force (IAF)** for **air defence of strategic assets**.

50. (c) The Subroto Cup was started in 1960.

It was started in memory of **Air Marshal Subroto Mukerjee**, who was the first Indian Chief of Air Staff.

Organized annually by the **Indian Air Force**, it encourages football at the grassroots level among school children.

Teams from **all over India and even abroad** have participated in this tournament.

51. (b) Raj Kumar Shukla, an indigo cultivator from Champaran, persuaded Mahatma Gandhi to come to Champaran to take up the cause of exploited indigo planters.

This eventually led to the **Champaran Satyagraha of 1917**, Gandhi's first satyagraha movement in India.

Indigo farmers in Champaran, Bihar, were forced to grow indigo on a portion of their land under the **Tinkathia system**.

52. (c) Statements 1 and 2 are correct.

DIKSHA, launched in **2017** is a **national platform developed by the Ministry of Education**.

It supports **school education** for teachers, students, and parents and hosts **e-content in multiple languages**.

It provides **QR-coded textbooks**, called **Energised Textbooks (ETBs)**, that link physical books with digital resources. Teachers and students across **states and UTs** can access them.

53. (c) Base-catalyzed haloform reaction involves a chain reaction mechanism that starts with a ketone group.

The haloform reaction specifically targets **methyl ketones**.

In base, the ketone forms an **enolate**, which undergoes **sequential α -halogenations** to give $R-CO-CBr_3/Cl_3$.

It's the classic test for methyl ketones.

54. (b) Statements 1 and 3 are correct.

Biomagnification is the progressive increase in the concentration of a toxic substance as it moves up successive trophic levels in a food chain.

Fat-soluble pollutants (lipophilic) are more likely to undergo biomagnification because they are stored in fatty tissues and not easily excreted. Water-soluble pollutants are excreted more easily.

Top carnivores (like birds of prey, humans, polar bears) are most affected since they are at the top of the food chain and accumulate the highest concentration of toxins.

55. (b) Cruise missile features best describes the Nirbhay missile.

Nirbhay is India's **long-range, all-weather, subsonic cruise missile** developed by **DRDO**.

Its range is around 1,000 km and is capable of carrying **nuclear or conventional warheads**.

It is **not** a ballistic missile, anti-tank missile, or surface-to-air missile.

56. (a) According to question,

$$6000 = \frac{P \times 5 \times 2}{100} \Rightarrow P = ₹60000$$

$$CI = 60000 \times \left[\left(1 + \frac{5}{100} \right)^2 - 1 \right]$$

$$= 60000 \times \left(\frac{441}{400} - 1 \right) = 60000 \times \frac{41}{400} = ₹6150$$